PV102 Day 4 Afternoon Lab

String Design Paper Lab

PV102 | CTS Day 20 | Afternoon Lab
String design follows limits, temperature, and shade reality. | Thursday, August 6, 2026

Mission

Teams compare three paper string options against controller limits and shade constraints.

Safety boundary

This lab is training work. Do not energize or connect equipment outside the instructor-approved procedure. When in doubt, stop, isolate, verify, and ask.

Scenario options

- Four 22 Vmp / 27 Voc training modules feeding a 100 V controller.
- Two shaded morning modules and two unshaded modules on a shed roof.
- Long ground-rack wire run where voltage drop and current both matter.

Procedure

- Read the module and controller cards.
- Build three possible string layouts using cards.
- Calculate approximate operating voltage/current and open-circuit voltage.
- Mark shade exposure from Day 2 scenario notes.
- Reject any option that violates a limit or ignores shade evidence.
- Handoff the safest first-pass string layout and its required verification checks.

Acceptance criteria

- Match the real job.
- Show the energy or work path.
- Name the dominant constraint.
- State the safety boundary.
- Record unknowns instead of guessing.
- Give a handoff another crew can use.

Team handoff form

Item	Response
Team members	
Scenario	
Primary decision	

Evidence	
Safety boundary	
Unknowns	
Next safest action	